

Executive Summary

This report presents two machine learning tasks. Task 1 addresses ten-class image classification on a subset of ImageNet. A baseline CNN with three convolutional blocks overfit severely averaging 59.34% +/- 0.63% validation accuracy across three seeds with a 38% training-to-validation gap. An improved model using Global Average Pooling, batch normalisation, dropout, and augmentation reached 70.27% +/- 1.61% with 90.1% fewer parameters. The gap between the two means comfortably exceeds both standard deviations, confirming the improvement is architectural. Eight out of ten classes improved, with chain saw and dog regressing. Task 2 addresses three-class sentiment classification on review data. Three approaches were compared: fine-tuned BERT-tiny, zero-shot BART-MNLI, and few-shot FLAN-T5. BART-MNLI achieved the strongest headline metrics (accuracy 0.869, macro F1 0.731), but bootstrap confidence intervals show the top three approaches are statistically tied. Only FLAN-T5's complete failure to predict the neutral class was reliably distinguishable. Prompt design had a substantial impact on zero-shot performance, and label noise from the 3-star mapping limits what any model can achieve.

Table of Contents

Executive Summary.....	1
<i>1. Task 1: Image Classification.....</i>	<i>3</i>
1.1 Introduction.....	3
1.2 Methodology.....	3
1.3 Experimental Setup.....	3
1.4 Results and Analysis.....	4
1.5 Discussion and Reflection.....	9
1.6 Conclusion.....	10
<i>Task 2: Sentiment Analysis on Review Data.....</i>	<i>11</i>
2.1 Introduction.....	11
2.2 Methodology.....	11
2.3 Experimental Setup.....	12
2.4 Results and Analysis.....	12
2.5 Discussion and Reflection.....	19
2.6 Conclusion.....	19
<i>References.....</i>	<i>21</i>

1. Task 1: Image Classification

1.1 Introduction

Image classification is a core problem within computer vision. The dataset used here is a subset of ImageNet (Deng et al., 2009) with ten categories: tench, dog, cassette player, chainsaw, church, French horn, bin lorry, petrol pump, golf ball and parachute. There are 9,469 training images and 3,925 validation images. The classes are roughly balanced but visually challenging, as several share cylindrical metallic features with considerable variation in lighting, pose, and background within each class. The aim was to build a baseline CNN, identify where it fails, and then design a second model to address those failures. The baseline was kept deliberately simple so that its weaknesses would be clearly visible rather than masked by regularisation.

1.2 Methodology

CNNs were used for this task. They use shared local filters, which are well-suited to natural images. The same edges and textures recur throughout various parts of an image, giving them far fewer parameters than a fully connected network on the same input (Krizhevsky et al., 2012). The baseline has three convolutional blocks (32, 64, 128 channels), ending with a flatten layer into a dense classifier. This is a standard starting point for image classification at this scale. The expectation was that it would overfit.

The improved model targets the baseline's weaknesses. The most important change replaces the flatten-to-dense block with Global Average Pooling. This collapses each feature map to a single value and removes the dense layer's memorisation capacity (Lin et al., 2013). Batch normalisation and dropout were added throughout. A fourth convolutional block at 256 channels compensates for the capacity removed from the head. Augmentation, such as random horizontal flips, small rotations, and zoom, expands the effective training set. This matters when the dataset is limited (Shorten and Khoshgoftaar, 2019). Adam was retained as the optimiser, with sparse categorical cross-entropy as the loss.

1.3 Experimental Setup

The dataset was loaded through Google Colab, and the models were implemented in TensorFlow. Class sizes range from 858 to 993 images, with a max-to-min ratio of 1.16, so accuracy is an appropriate headline metric. Images were loaded at 144×144 in batches of 32, with caching and prefetching to maintain throughput. Sample images were inspected to verify that labels were correctly assigned (**Figure 1**). Training and validation accuracy and loss were tracked at every epoch, with confusion matrices and per-class precision, recall, and F1 for fine-grained error analysis.

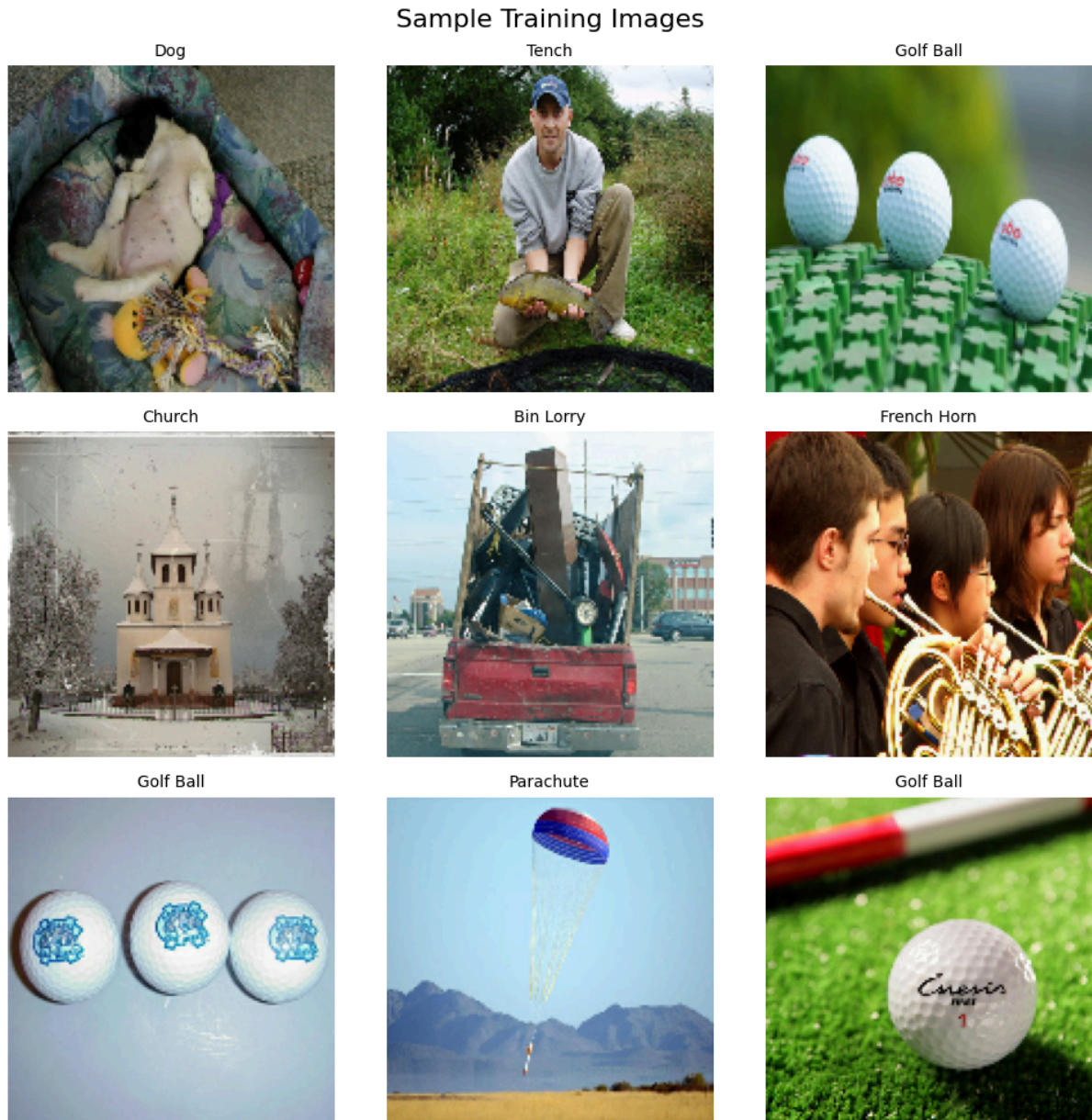


Figure 1. Sample training images with class labels, used to verify correct dataset loading.

The baseline was trained for ten epochs with no early stopping, consistent with its function as a deliberately uncontrolled reference. The improved model was trained for up to 20 epochs with early stopping on the validation loss (patience = 5), ReduceLROnPlateau, and ModelCheckpoint. Dropout was set to 0.4 in the deeper feature stage and 0.3 in the classifier head. To quantify run-to-run variance, both models were trained across three random seeds (42, 123, 2024). Reported headline figures are the mean and standard deviation across seeds rather than single-run values.

1.4 Results and Analysis

The baseline overfitted substantially (**Figure 2**). Across three seeds, final-epoch training accuracy averaged 97.6% while validation accuracy averaged just 59.34% +/- 0.63%, an

overfitting gap of roughly 38 percentage points. Validation loss climbed steadily from epoch 2 onwards in all three runs while training loss collapsed below 0.1, the classic memorisation signature. The cause is architectural: 4,194,432 of the baseline’s 4,288,970 parameters (97.8%) sit in the dense layer immediately after the flatten, roughly 440 parameters per training image (Bejani and Ghatee, 2021). The small standard deviation across seeds (0.63%) confirms this is a stable failure mode rather than something seed-dependent.

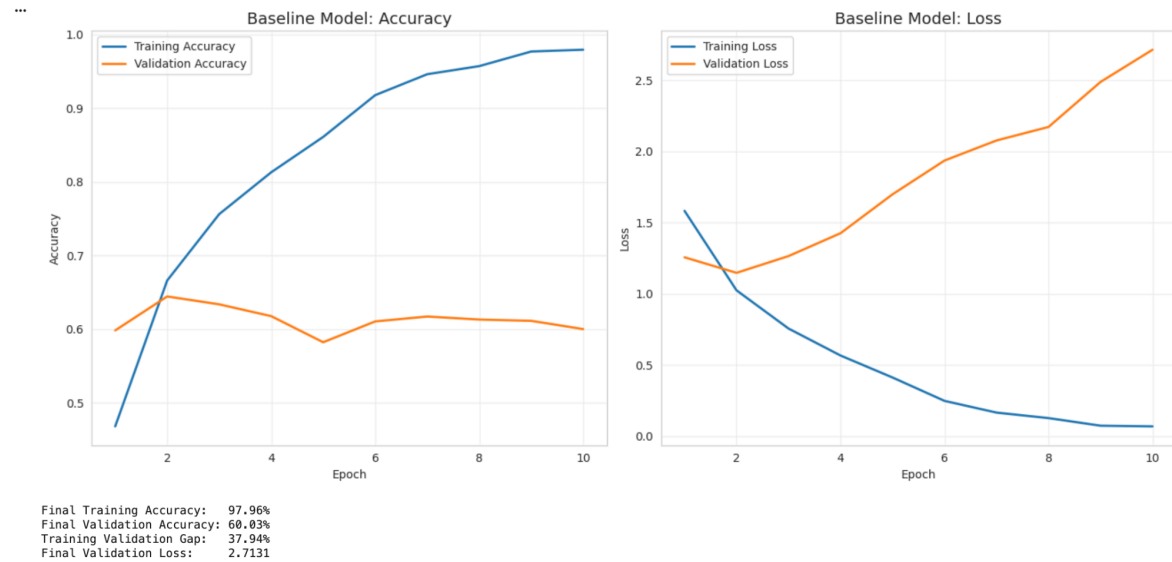


Figure 2. Baseline training curves at 10 epochs.

The confusion matrix (**Figure 3**) shows that the damage was uneven across classes. The petrol pump scored highest at 73.27%, while the chain saw managed only 41.71% and the cassette player 43.70%. Chain saw’s errors spread across multiple classes, with 29 instances misclassified as dog, 40 as French horn, and 73 as petrol pump. The pattern suggests a model that has picked up broad visual cues (fur, metallic surfaces, cylindrical shapes) without learning what genuinely distinguishes the classes.

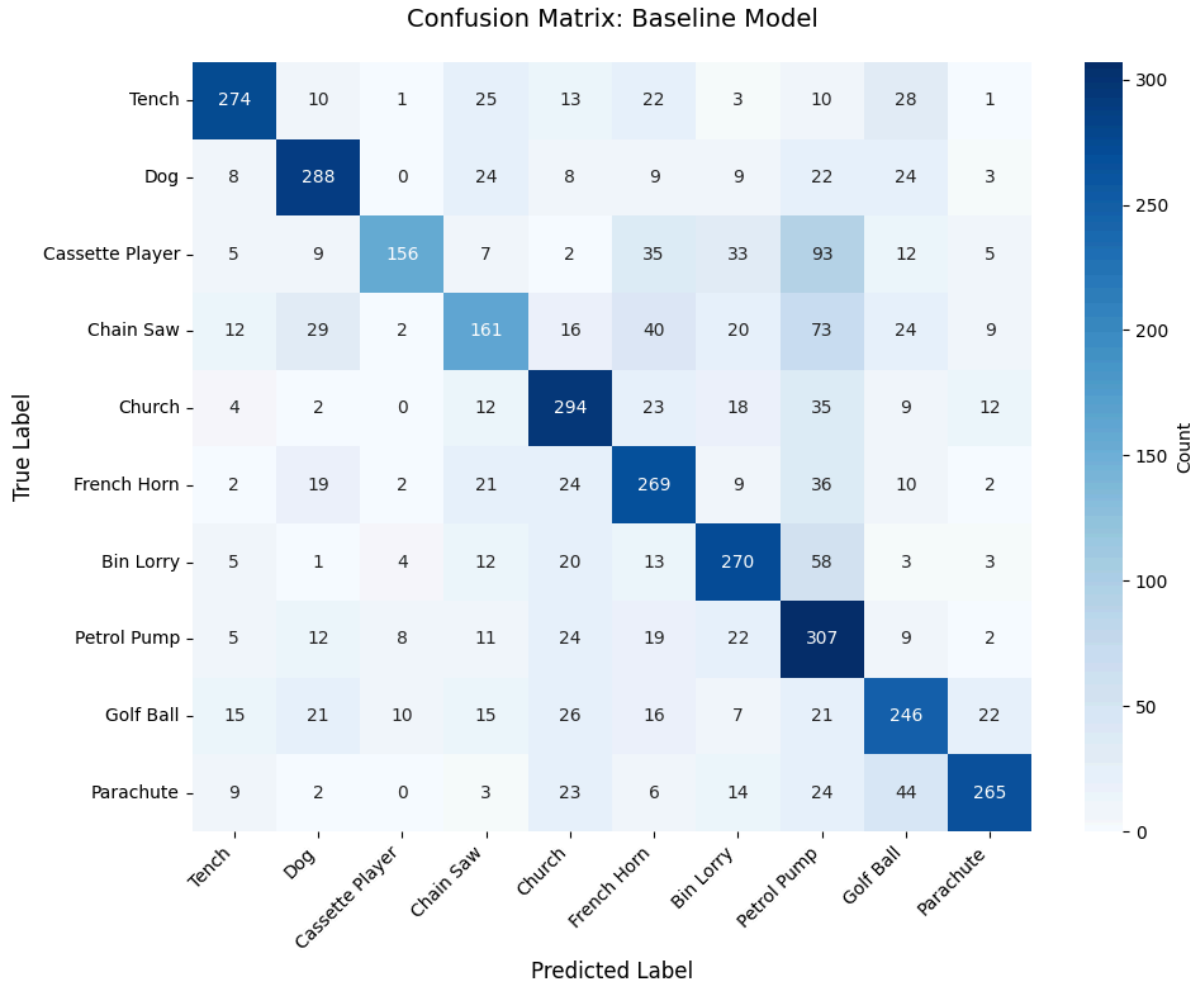
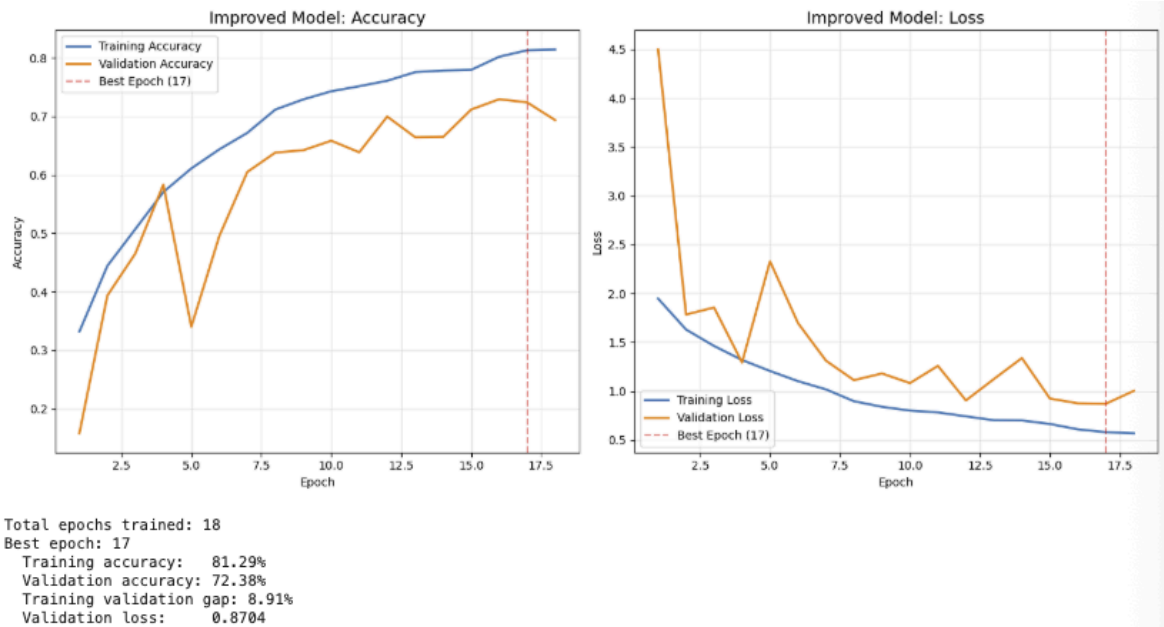


Figure 3. Baseline confusion matrix on the validation set.

The improved model achieved 70.27% +/- 1.61% validation accuracy on three seeds. It used 90.1% fewer parameters and had a much narrower overfitting gap (**Table 1**). The best epoch's training accuracy was lower than the baseline's, but that's the point: lower training accuracy shows the model is generalising rather than memorising. Learning continued past epoch 14 before early stopping ended training (Figure 4). The gap between the improved mean and the baseline mean (70.27% vs 59.34%) easily exceeds both standard deviations, confirming the improvement is architectural, not due to luck. The improved model's standard deviation was larger than the baseline's (1.61% vs 0.63%), suggesting the deeper architecture with dropout and augmentation is more sensitive to weight initialisation.

Table 1. Baseline vs Improved Model: multi-seed results.

Metric	Baseline (3 seeds)	Improved (3 seeds)
Training Accuracy	97.6%	79.93%
Validation Accuracy	59.34% +/- 0.63%	70.27% +/- 1.61%
Overfitting Gap	38.31 +/- 0.54%	9.67% +/- 1.97%
Validation Loss	2.71	0.98
Total Parameters	4,288,970	424,522 (90.1% reduction)

**Figure 4.** Improved model training curves with best epoch marked.

Per-class accuracy (**Figure 5**) improved on eight of ten classes. The largest gains went to classes the baseline struggled with most: cassette player nearly doubled from 43.70% to 82.91%, golf ball rose from 61.65% to 85.46%, and parachute jumped from 67.95% to 91.28%. This is consistent with Global Average Pooling acting as a structural regulariser, forcing the convolutional layers to learn more discriminative features rather than relying on the dense head to memorise them (Lin et al., 2013). The improved model's confusion matrix (**Figure 6**) confirms the shift, with cassette player correct predictions rising from 156 to 296. Three classes regressed: dog dropped from 72.91% to 56.96%, chain saw fell further from 41.71% to 30.57%, and petrol pump slipped from 73.27% to 68.74%. Chain saw remained the weakest class in both models, with its errors in the improved model spreading widely across petrol pump, cassette player, golf ball, and parachute. The confusion matrix shows no single dominant misclassification, suggesting the model still struggles to distinguish mechanical cylindrical objects at 144 x 144 resolution. The chain saw regression is particularly informative. The same regularisation that improved most classes appears to have removed exactly the kind of surface-level pattern matching that the baseline relied on for

chain saw. This suggests that visually ambiguous classes may need targeted intervention rather than general-purpose regularisation.

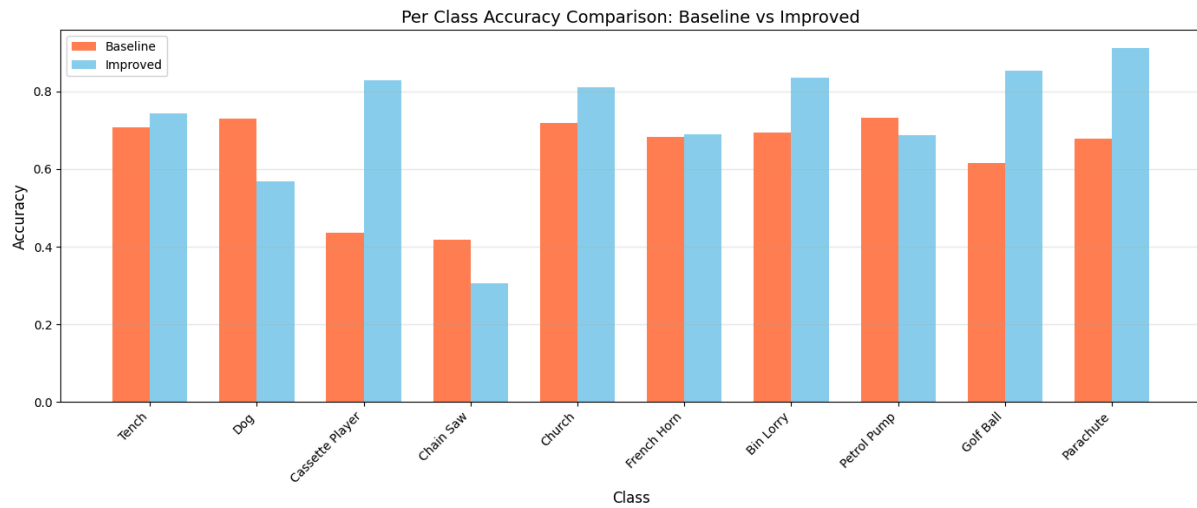


Figure 5. Per-class accuracy comparison: baseline vs improved.

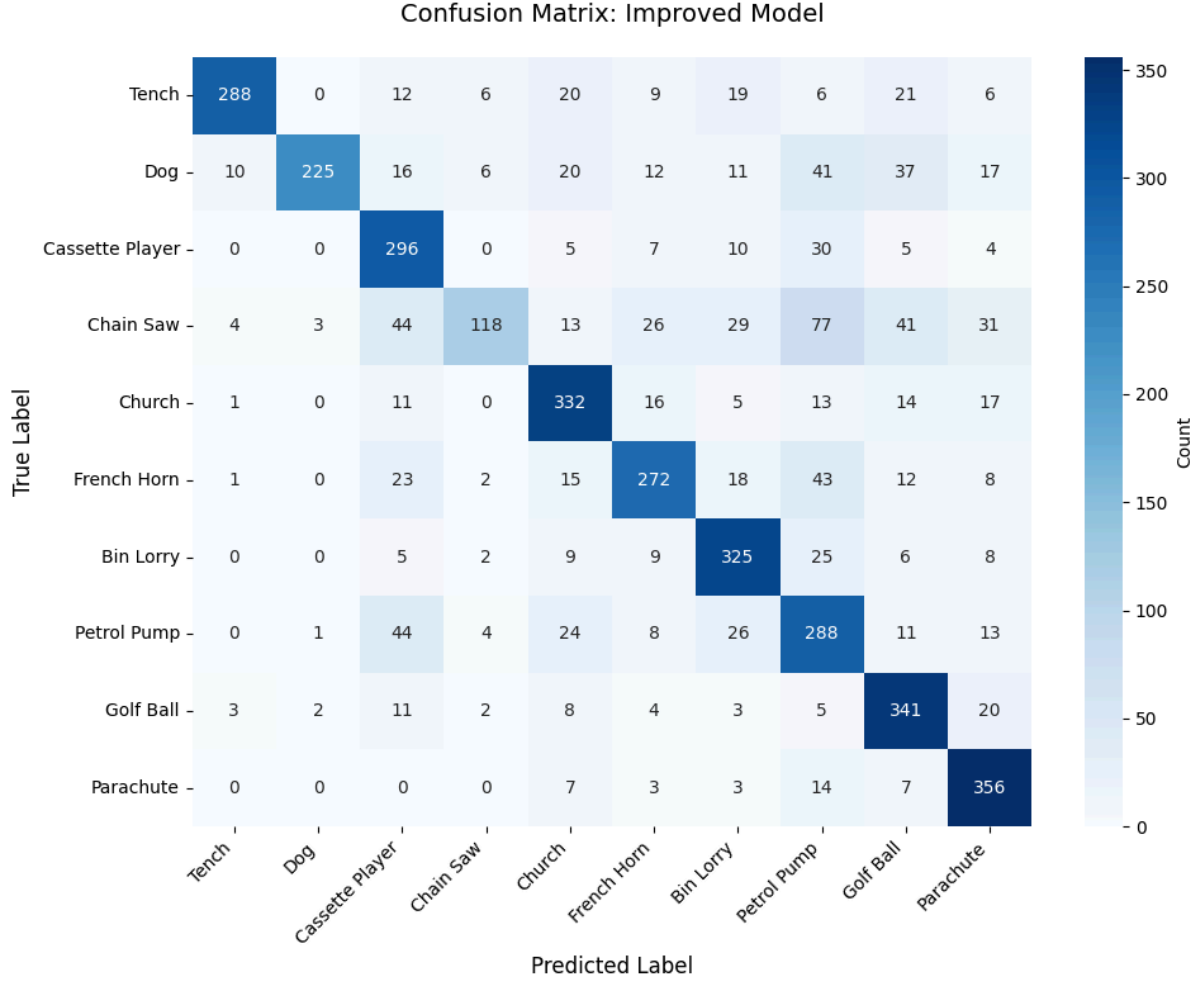


Figure 6. Improved model confusion matrix on the validation set.

1.5 Discussion and Reflection

The most fundamental assumption here is that training a CNN from scratch is appropriate for a dataset of this size. Transfer learning would almost certainly perform better, since pretraining on ImageNet captures low-level features that any natural-image task can reuse (Shorten and Khoshgoftaar, 2019). A second issue is the 144×144 input resolution, which may discard detail that matters for the harder classes. Prior work (Shorten and Khoshgoftaar, 2019) reports that models trained at 256×256 and 512×512 reached top-5 errors of 7.96% and 7.42%, respectively, with their ensemble at 6.97%, suggesting that different resolutions can act as augmentation. Chainsaw remains the weakest class, and a more specialised augmentation might help. SamplePairing, which mixes two images by averaging RGB pixel values, reduced error from 43.1% to 31.0% on a 1,000-sample CIFAR-10 subset (Shorten and Khoshgoftaar, 2019).

No separate test set was held back from model selection. Early stopping, checkpointing, and final metrics all used the same validation split. The reported figures, therefore, overstate performance on unseen data. A proper three-way split into train, validation, and test sets would allow early stopping and checkpoint selection on the validation set while reserving the test set for a single final evaluation, giving more honest headline numbers. Beyond the

evaluation setup, using a pretrained backbone such as VGG or ResNet (Simonyan and Zisserman, 2014) would be the next step if repeating this work.

1.6 Conclusion

This task compared a baseline CNN with an improved version on a ten-class image classification problem. Across three seeds, the baseline averaged 59.34% $\pm$ 0.63% validation accuracy. The improved model reached 70.27% $\pm$ 1.61%, with 90.1% fewer parameters. The gap between the two means comfortably exceeds both standard deviations. This confirms that the architectural changes drove the improvement. Eight out of ten classes improved, with the largest gains in those that struggled most at baseline. Chain saw’s regression highlights a limitation of general regularisation for visually ambiguous classes, where augmentation or higher resolution may be needed. The principal lesson is that matching model capacity to the dataset and directly resisting memorisation with architecture achieved more here than any single regularisation technique.

Task 2: Sentiment Analysis on Review Data

2.1 Introduction

This task addressed three-class sentiment classification on review data. Star ratings were mapped to labels: 1-2 negative, 3 neutral, 4-5 positive. The class distribution is heavily skewed: 67% positive, 23% negative, and 10% neutral (**Figure 7**). Three models were evaluated on the same 5,000-example test set: fine-tuned BERT-tiny (supervised baseline), zero-shot BART-MNLI (hypothesis-based classification), and few-shot FLAN-T5 (in-context learning). Macro F1 is the primary metric, as it treats all classes equally. Threshold tuning for BERT-tiny and prompt-sensitivity probes on prompt-based models was also performed.



Figure 7. Train and test class distributions.

2.2 Methodology

The three models cover the main approaches to modern NLP classification. BERT-tiny is the supervised baseline: a two-layer bidirectional transformer with 4.4 million parameters (Turc et al., 2019). It is fine-tuned with inverse-frequency class weighting, so errors on the rare neutral class incur a heavier penalty. BART-MNLI is the zero-shot model: a 12-layer encoder-decoder (400 million parameters), pretrained via denoising and fine-tuned on MultiNLI (Lewis et al., 2020; Williams et al., 2018). Each label is expressed as a hypothesis. The model assigns each review to the label with the highest entailment score. FLAN-T5 is the few-shot model: an instruction-tuned T5 variant (250 million parameters), trained on 1,836 tasks (Chung et al., 2024). Six demonstration examples (two per class) are placed in the prompt before each query.

2.3 Experimental Setup

The dataset contains approximately 1 GB of training data and 103 MB of test data in JSONL format. BERT-tiny was fine-tuned on a 100,000-example random subsample, which preserved the 67/23/10 class proportions without specific stratification. All three models were evaluated on the same 5,000 example test subset. Macro F1 is the primary metric, with weighted F1, the confusion matrix, and per-class precision and recall reported alongside. BERT-tiny was trained with Adam at a learning rate of $2e-5$, batch size 64, and class weights of approximately 6.7x for neutral, for up to 8 epochs with early stopping (patience = 2). BART-MNLI was tested under three hypothesis templates (Table 2). FLAN-T5 used six balanced demonstrations with three diagnostic probes (Table 3) on 50 examples each.

2.4 Results and Analysis

BERT-tiny validation accuracy peaked at 80% on epoch 7 with weighted F1 0.826 and macro F1 0.71 (**Figures 8-9**). Per-class F1 was 0.81 negative, 0.43 neutral, and 0.89 positive. Neutral had high recall (0.69) but low precision (0.31), as the class weighting pushed the model towards predicting neutral for ambiguous cases (**Figure 10**). Of the 1,001 errors, 503 (50%) were positive reviews predicted as neutral, with three patterns recurring: praise without intensifiers ('good', 'fine'), reviews with a complaint followed by positive resolution, and positive events described in factual language. The two-layer architecture appears too small to distinguish moderate positive sentiment from genuine neutrality.

```
Epoch 1/8
1563/1563 [=====] - 140s 70ms/step - loss: 0.8098 - accuracy: 0.6981 - val_loss: 0.5685 - val_accuracy: 0.7684
Epoch 2/8
1563/1563 [=====] - 61s 39ms/step - loss: 0.6327 - accuracy: 0.7604 - val_loss: 0.5255 - val_accuracy: 0.7744
Epoch 3/8
1563/1563 [=====] - 59s 38ms/step - loss: 0.5911 - accuracy: 0.7744 - val_loss: 0.4967 - val_accuracy: 0.7890
Epoch 4/8
1563/1563 [=====] - 59s 38ms/step - loss: 0.5647 - accuracy: 0.7862 - val_loss: 0.4957 - val_accuracy: 0.7886
Epoch 5/8
1563/1563 [=====] - 59s 38ms/step - loss: 0.5404 - accuracy: 0.7958 - val_loss: 0.4763 - val_accuracy: 0.7994
Epoch 6/8
1563/1563 [=====] - 60s 38ms/step - loss: 0.5205 - accuracy: 0.8033 - val_loss: 0.4893 - val_accuracy: 0.7920
Epoch 7/8
1563/1563 [=====] - 60s 38ms/step - loss: 0.5020 - accuracy: 0.8101 - val_loss: 0.4760 - val_accuracy: 0.7998
Epoch 8/8
1563/1563 [=====] - 59s 38ms/step - loss: 0.4844 - accuracy: 0.8177 - val_loss: 0.4875 - val_accuracy: 0.7924
```

Figure 8. BERT-tiny training trajectory.

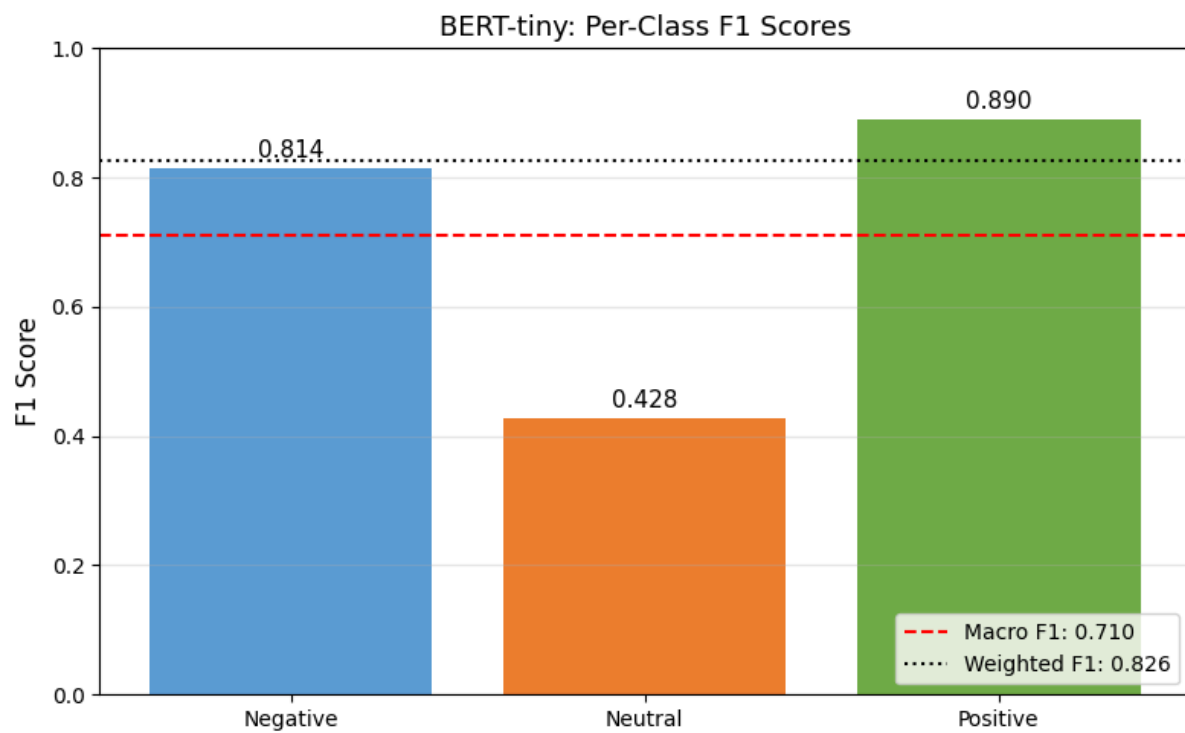


Figure 9. BERT- tiny per class metrics.

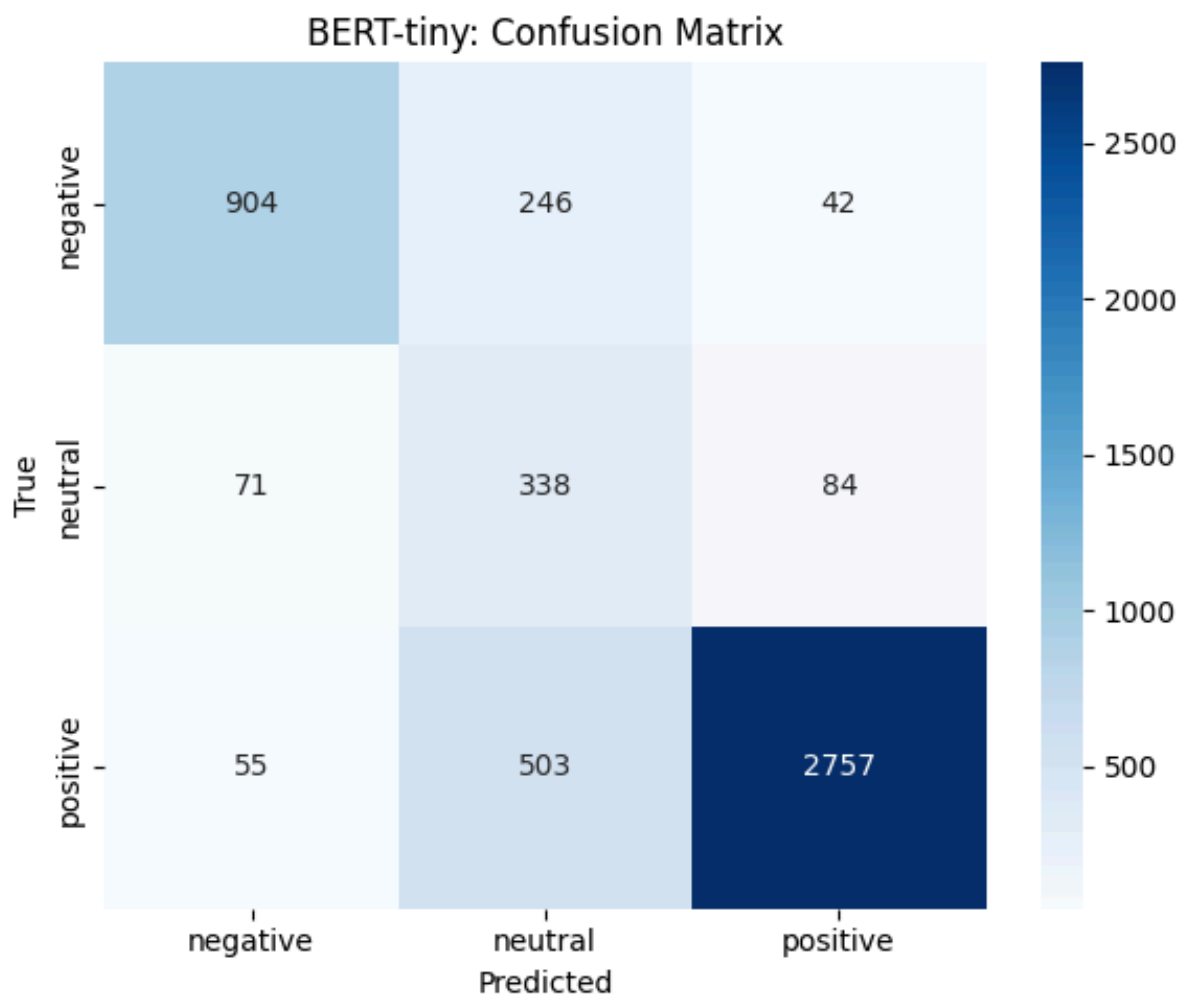


Figure 10. BERT-tiny confusion matrix at default argmax.

Sweeping the neutral threshold from 0.3 to 0.7 in 0.05 steps, macro F1 peaked at $\tau = 0.65$ (**Figure 11**). Accuracy went from 0.8 to 0.833, macro F1 from 0.71 to 0.722. Neutral F1 stayed at 0.43 (**Figure 12**): precision rose from 0.31 to 0.37, but recall dropped from 0.69 to 0.51. Threshold tuning, therefore, redistributes errors without addressing the underlying issue, indicating a representational rather than a calibration problem. The threshold maximising neutral F1 is also 0.55, not 0.65, so optimising for one metric is not the optimum for all. Tuning was done on the test set, which technically inflates the gain (Cawley and Talbot, 2010).

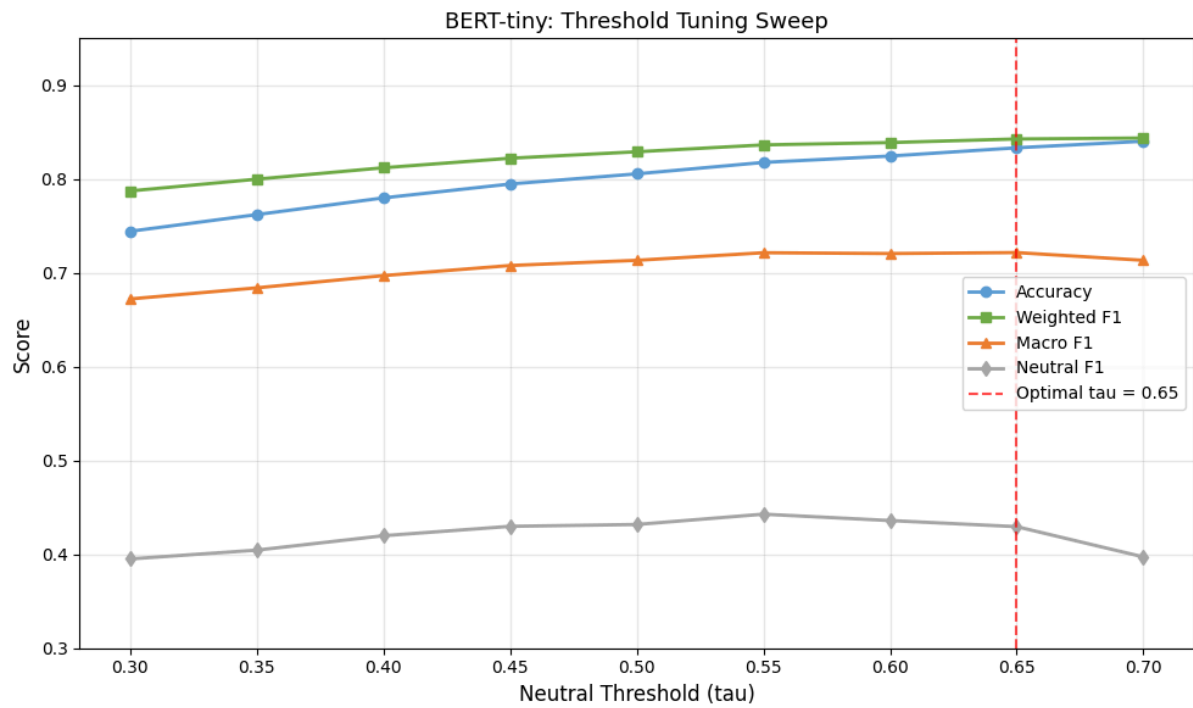


Figure 11. Threshold sweep; macro F1 is optimised at $\tau = 0.65$.

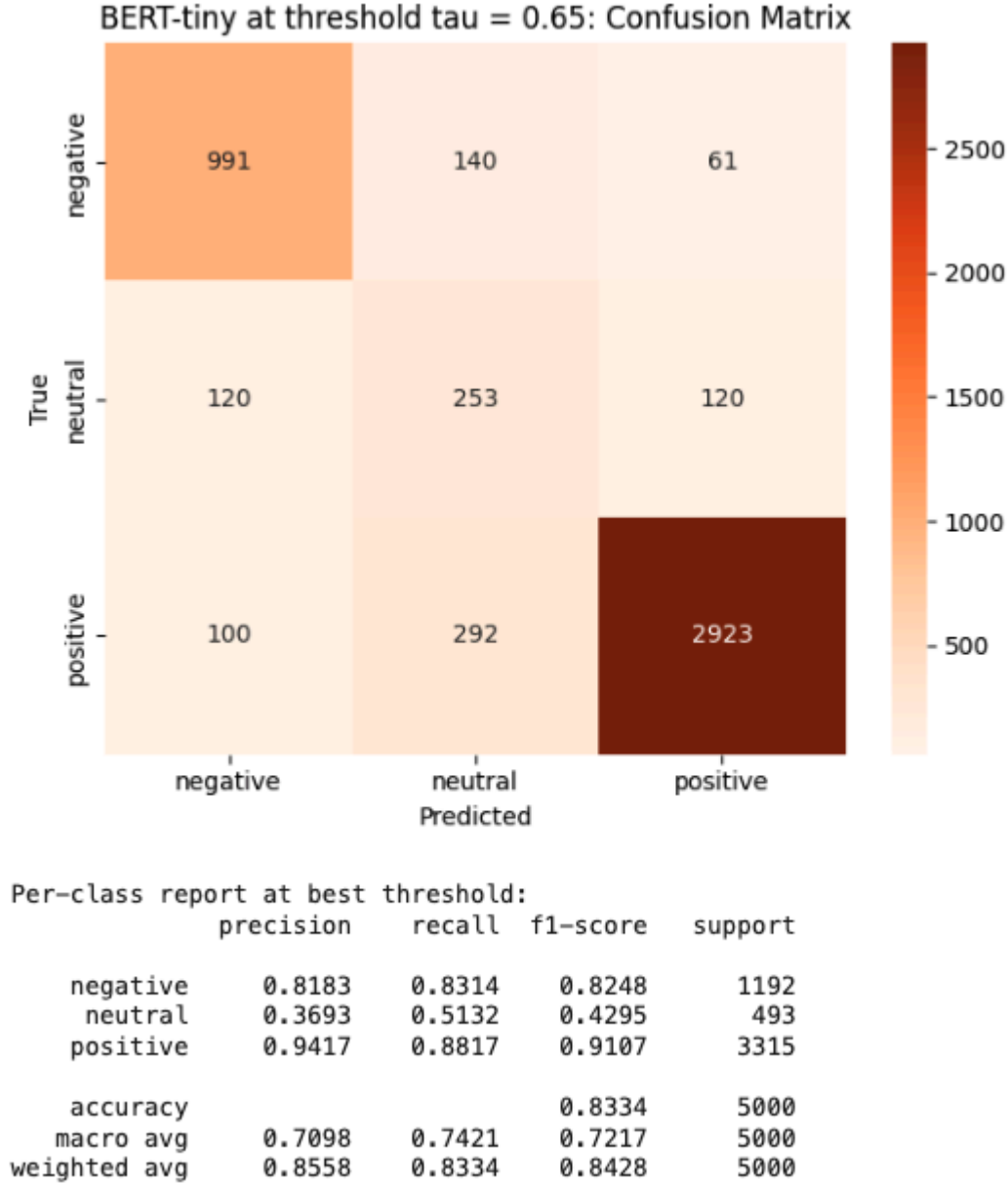


Figure 12. BERT-tiny confusion matrix at tau = 0.65.

BART-MNLI achieved the strongest headline metrics, with an accuracy of 0.869 and a macro F1 of 0.731. Negative recall reached 0.91, and positive recall reached 0.93. Neutral F1 was 0.4, slightly below threshold-tuned BERT-tiny (0.43), despite BART having approximately 100x more parameters. Figure 13 shows that the failure mode of BERT-tiny is the inverse of BART: BART under-predicts neutral, while BERT-tiny over-predicts it. Prompt sensitivity was substantial. Few-shot prompts are known to be sensitive to demonstration order (Lu et al., 2022); the equivalent for zero-shot is whether label phrasing has a comparable effect. Under bare label words, macro F1 collapsed from 0.751 to 0.594 and neutral F1 from 0.449 to 0.039 (Table 2). The emotion-framed variant improved both (macro F1: 0.758; neutral F1: 0.488).

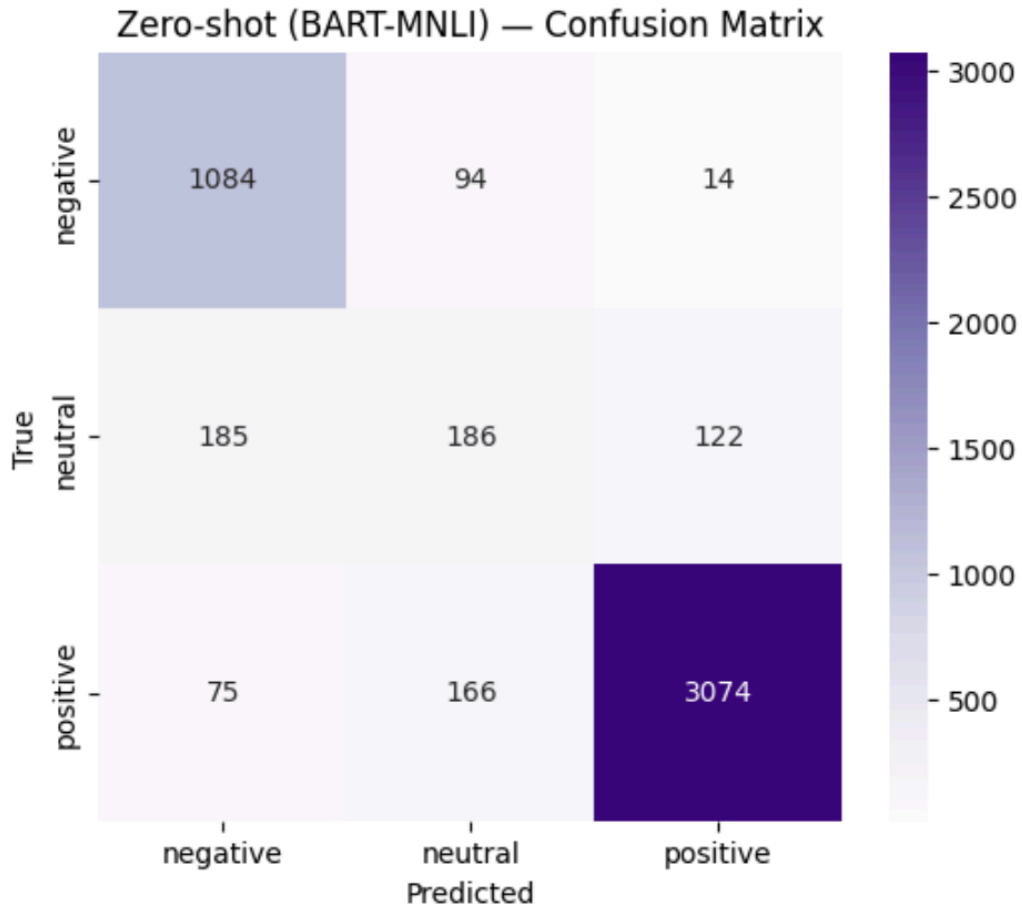


Figure 13. BART-MNLI confusion matrix.

Variant	Accuracy	Weight. F1	Macro F1	Neutral F1
Sentence (original)	0.882	0.879	0.751	0.449
Single word	0.852	0.815	0.594	0.039
Emotion-framed	0.858	0.870	0.758	0.488

Table 2. BART-MNLI prompt sensitivity (n = 500).

Few-shot prompting with FLAN-T5 reached 0.831 accuracy but the lowest macro F1 (0.564), because the model predicted neutral exactly zero times across all 5,000 examples (Figure 14). Three probes (**Table 3**, **Figure 15**) ruled out the obvious explanations: relabelling neutral as ‘mixed’, numeric labels, and a neutral-heavy demonstration set all produced near-deterministic binary outputs. In-context learning is known to tolerate even random labels (Min et al., 2022), so the complete absence of the class suggests that instruction-tuning priors override the demonstrations entirely.

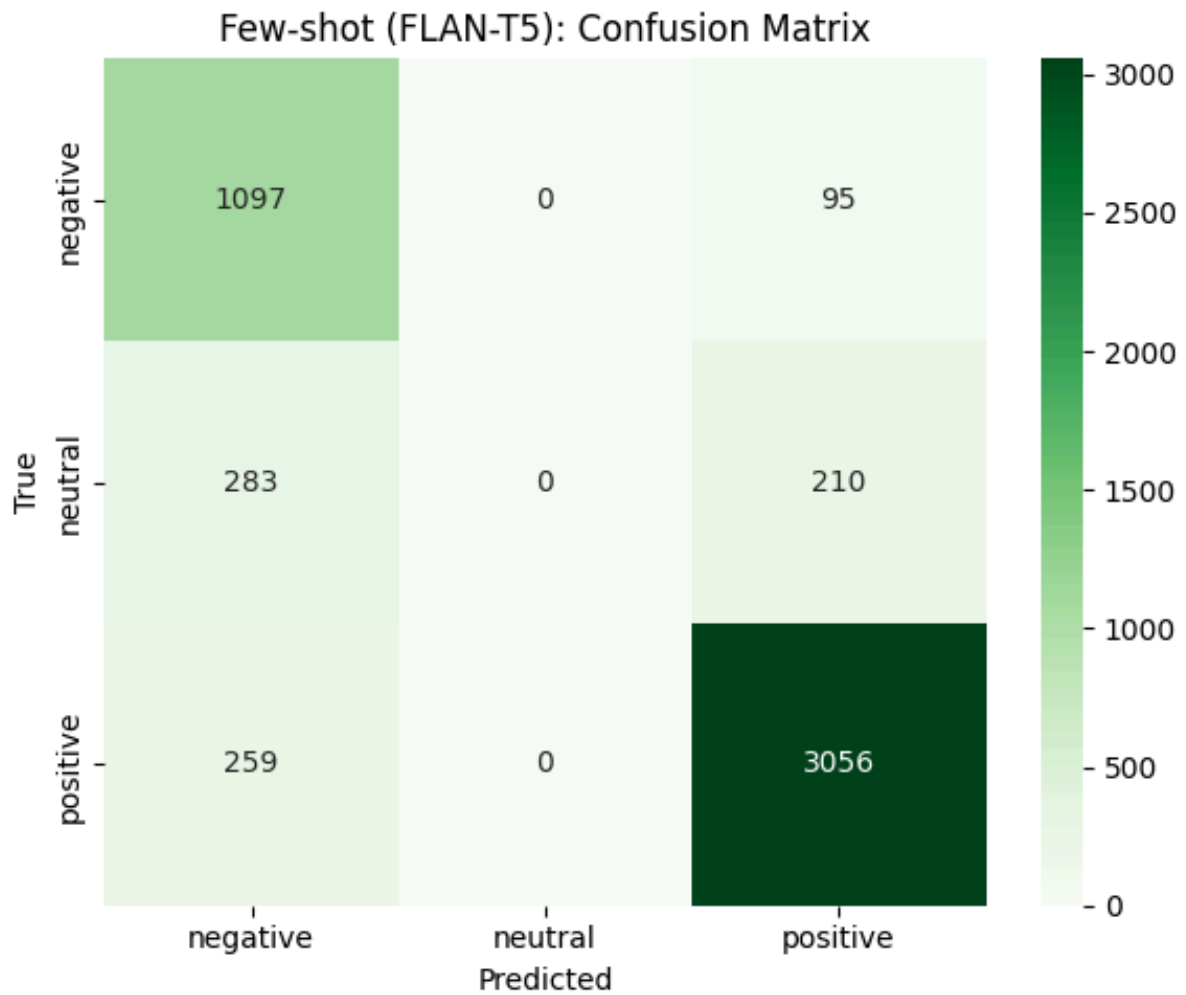


Figure 14. FLAN- T5 confusion matrix.

Probe	Manipulation	Output (n=50)
1	Relabel “neutral” as “mixed”	33 positive, 17 neutral, 0 mixed
2	Numeric labels (1,2 3)	36 = “3” 14 = “1”. 0 = “2”
3	4 neutral demos out of 6	33 positive, 17 negative, 0 neutral

Table 3. FLAN-T5 prompt-sensitivity probes (n = 50).

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Probe 1 (neutral as 'mixed'): Counter({'positive': 33, 'negative': 17})
Probe 2 (numeric labels): Counter({'3': 36, '1': 14})
Probe 3 (neutral-heavy demonstrations): Counter({'positive': 33, 'negative': 17})

```

Figure 15. Probe counter outputs.

Table 4 compares all four configurations with bootstrap confidence intervals. The CIs for BERT-tiny argmax, BERT-tiny tuned, and BART-MNLI overlap, so the apparent ranking among them reflects statistical noise rather than a real ordering. Only FLAN-T5 clearly sits outside this group; the 0.021 macro-F1 gap between BERT-tiny argmax and BART-MNLI does not hold up under bootstrapping.

Model	Accuracy	Weighted F1	Macro F1	Neutral F1	95% Confidence Interval (CI)
BERT-tiny (argmax)	0.800	0.826	0.710	0.43	[0.6950, 0.7241]
BERT-tiny (tau = 0.65)	0.833	0.843	0.722	0.43	[0.7053, 0.7373]
Zero-shot (BART-MNLI)	0.869	0.868	0.731	0.40	[0.7150, 0.7473]
Few-shot (FLAN-T5)	0.831	0.792	0.564	0.00	[0.5561, 0.5706]

Table 4. Final model comparison with 95% CI.

2.5 Discussion and Reflection

BART-MNLI's notable performance is unsurprising given that its MultiNLI pretraining maps almost directly onto the entailment task that zero-shot reduces to. The ranking shifts depending on the metric: few-shot beats argmax BERT-tiny by 0.031 in accuracy but trails by 0.146 in macro F1, reflecting how much each model is affected by the imbalance.

The trade-offs go beyond accuracy. BART's 400 million parameters make inference considerably more expensive than BERT-tiny's 4.4 million, so in production, the supervised baseline would be cheaper to scale. A deeper problem lies in the labels themselves: half of BERT-tiny's errors were positive reviews predicted as neutral, indicating that many 3-star reviews carry mild positive sentiment rather than indifference. Any model trained on this mapping inherits that noise.

The two prompt-based models also responded differently to the intervention. BART's neutral performance shifted substantially with the phrasing of the hypothesis, suggesting the problem is partly fixable through prompt design. FLAN-T5 was unaffected by any of the three probes, suggesting that instruction-tuning priors, rather than prompt engineering, were the limiting factor. The probes tested only label phrasing, not demonstration order, which is a known source of variance (Lu et al., 2022) and would justify further investigation.

A more robust pipeline would tune the threshold on a held-out validation set, conduct a more systematic prompt search, and potentially ensemble supervised and prompt-based outputs. The findings here are based on a single 5,000 review test subset and do not capture variation across review categories or rating scales.

2.6 Conclusion

BART-MNLI achieved the strongest headline metrics, but the bootstrap analysis showed the top three approaches are statistically similar. Only FLAN-T5's collapse on the neutral class was reliably distinguishable. Threshold tuning improved overall accuracy but not

minority-class performance, and label noise from the 3-star mapping further constrains what any model can achieve here. Three takeaways: zero-shot models perform well in data-scarce settings when accuracy matters, supervised models with calibration are more reliable for less-represented classes, and prompt-based methods need validation before any single result can be trusted.

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